Verilog Live Demo

COA Laboratory

xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/vivado-design-tools/2020-2.html

Solutions Products Support XILINX

/ Support / Downloads

Downloads

Licensing Help | Alveo Accelerator Card

Vivado (HW Developer) | Vitis (SW Developer) | Vitis Embedded Platforms | Alveo Packages | PetaLinux | Device Models | Documentation Navigator

Version

- 2020.2
- 2020.1
- 2019.2
- Vivado Archive
- ISE Archive
- CAE Vendor Libraries Archive

Vivado Design Suite - HLx Editions - 2020.2 Full Product Installation

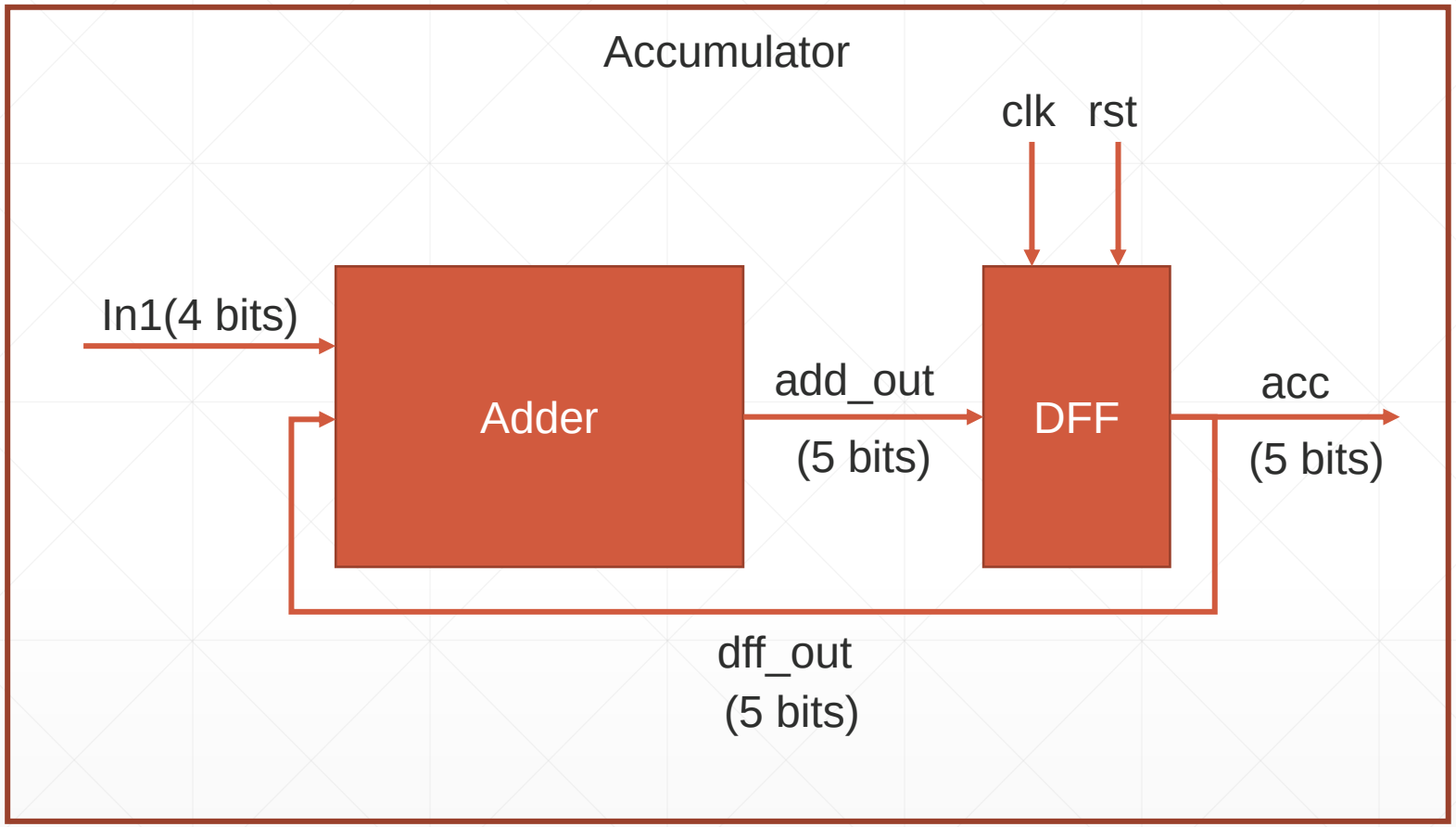
Important Information

Vivado Design Suite 2020.2 is now available

- Public access support for the Xilinx Versal Platforms
- Petalinux now a part of Xilinx Unified Installer
- Access Block Design container now to create team-based designs
- Abstract Shell for Dynamic Function eXchange
- 2020.2 Introduces Vitis HLS for Vivado flows
- Add-on for MATLAB and Simulink (Unified Model Composer and System Generator)

Download Includes	Vivado Design Suite HLx Editions (All Editions)
Download Type	Full Product Installation
Last Updated	Nov 24, 2020
Answers	2020.x - Vivado Known Issues
Documentation	Release Notes OS Support Update What's New in Vivado
Support Forums	Installation and Licensing

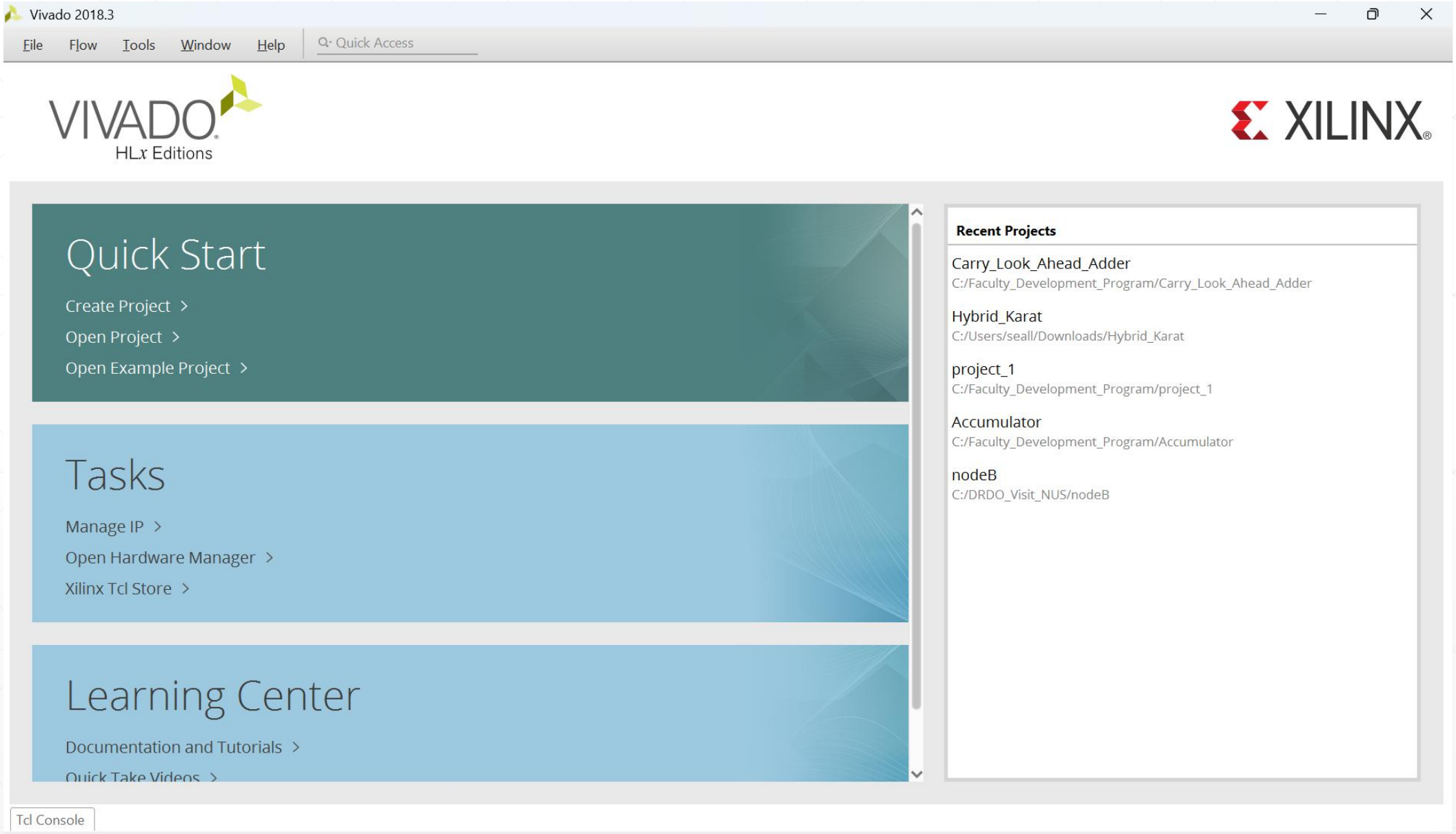
https://www.xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/vivado-design-tools/2020-2.html


Accumulator
clk
rst
In1(4 bits)
Adder
add_out
(5 bits)
DFF
acc
(5 bits)
dff_out
(5 bits)

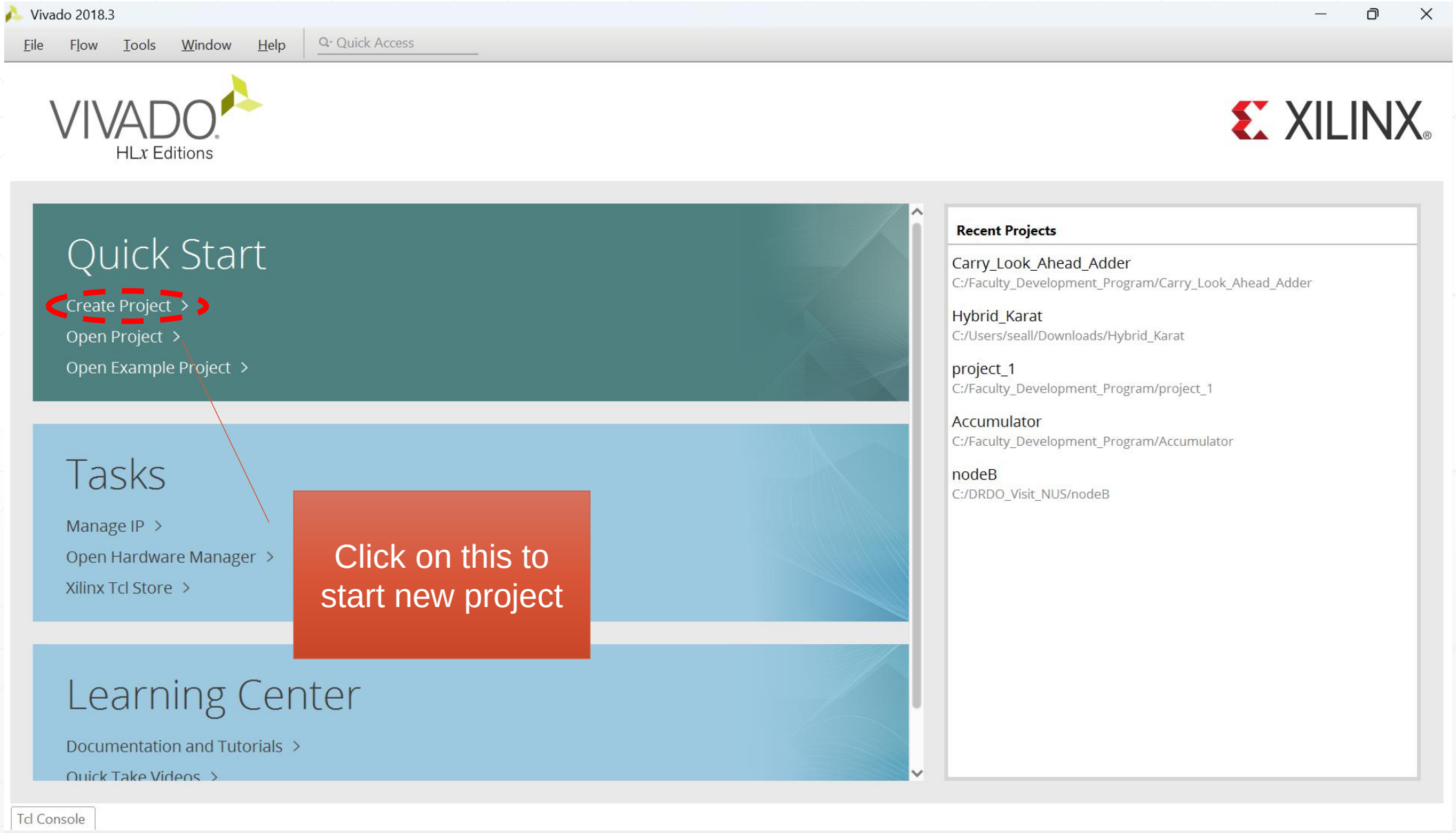
Vivado Tutorial

COA Laboratory

Start Screen of VIVADO

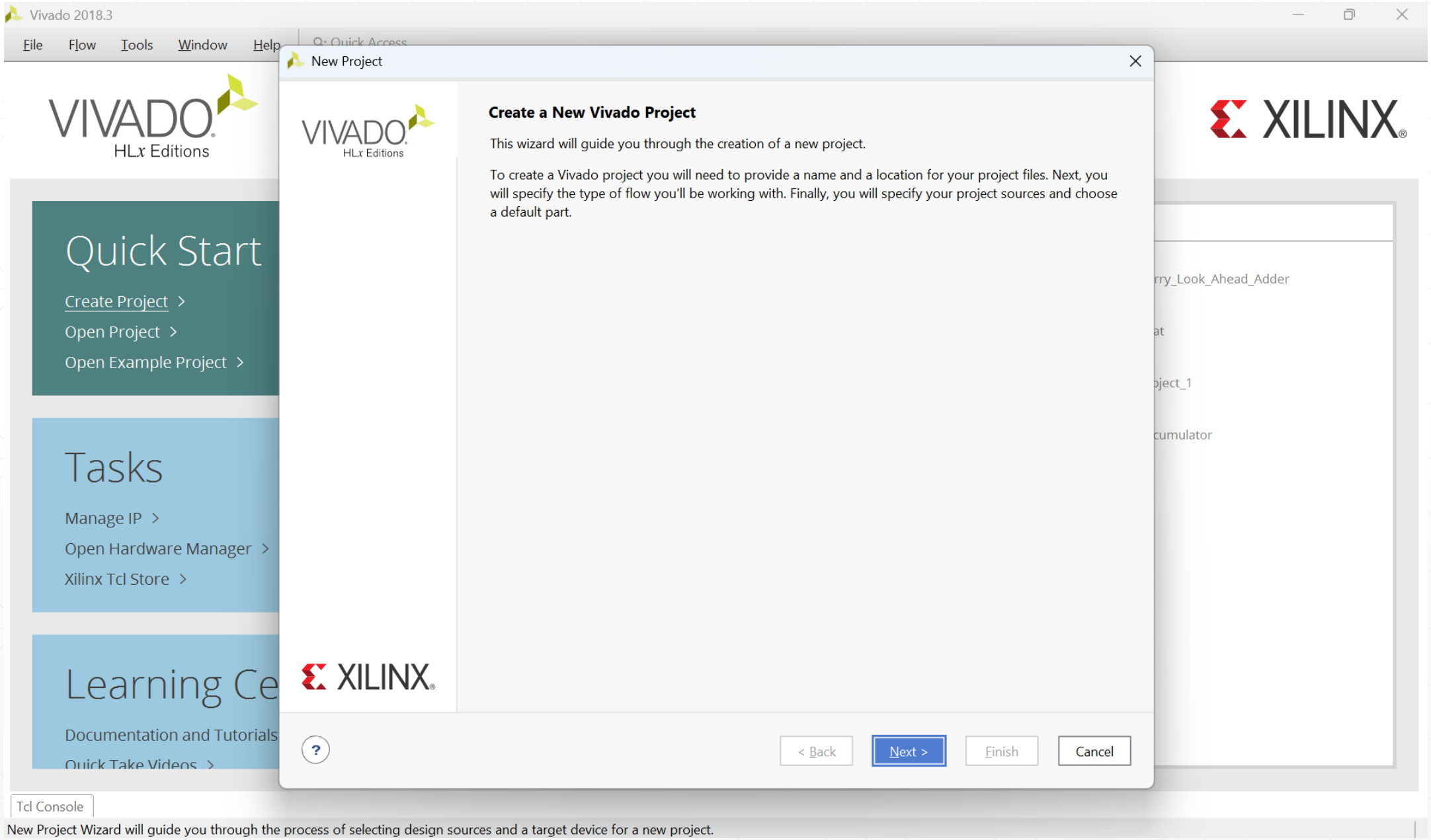


Start Screen of VIVADO



Click on this to start new project

Click on Next



Vivado 2018.3

File Flow Tools Window Help Q: Quick Access

VIVADO HLx Editions

XILINX

Quick Start

Create Project >

Open Project >

Open Example Project >

Tasks

Manage IP >

Open Hardware Manager >

Xilinx Tcl Store >

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New Project

Project Name

Enter a name for your project and specify a directory where the project data files will be stored.

Project name: project_1

Project location: C:/Users/seall/Downloads/Hybrid_Karat

Create project subdirectory

Project will be created at: C:/Users/seall/Downloads/Hybrid_Karat/project_1

Click on Next

< Back | Next > | Finish | Cancel

Tcl Console

New Project Wizard will guide you through the process of selecting design sources and a target device for a new project.

Vivado 2018.3

File Flow Tools Window Help Q: Quick Access

VIVADO HLx Editions

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oject_1
cumulator

New Project

Project Type

Specify the type of project to create.

- (●) RTL Project
  You will be able to add sources, create block designs in IP Integrator, generate IP, run RTL analysis, synthesis, implementation, design planning and analysis.
  - [ ] Do not specify sources at this time
- ( ) Post-synthesis Project: You will be able to add sources, view device resources, run design analysis, planning and implementation.
  - [ ] Do not specify sources at this time
- ( ) I/O Planning Project
  Do not specify design sources. You will be able to view part/package resources.
- ( ) Imported Project
  Create a Vivado project from a Synplify, XST or ISE Project File.
- ( ) Example Project
  Create a new Vivado project from a predefined template.

Click on Next

? < Back Next > Finish Cancel

Tcl Console

New Project Wizard will guide you through the process of selecting design sources and a target device for a new project.

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File Flow Tools Window Help Quick Access

VIVADO HLx Editions

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oject_1

cumulator

New Project

Add Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those files, to add to your project. Create a new source file on disk and add it to your project. You can also add and create sources later.

Use Add Files, Add Directories or Create File buttons below

Add Files | Add Directories

Click on Next

Scan and add RTL include files into project

Copy sources into project

Add sources from subdirectories

Target language: Verilog Simulator language: Mixed

< Back | Next > | Finish | Cancel

Tcl Console

New Project Wizard will guide you through the process of selecting design sources and a target device for a new project.

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oject_1
cumulator

New Project

Add Constraints (optional)

Specify or create constraint files for physical and timing constraints.

Use Add Files or Create File buttons below

Click on Next

Add Files | Create File

Copy constraints files into project

? | < Back | Next > | Finish | Cancel

Tcl Console

New Project Wizard will guide you through the process of selecting design sources and a target device for a new project.

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File Flow Tools Window Help Quick Access

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XILINX

Quick Start

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Tasks

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New Project

Default Part

Choose a default Xilinx part or board for your project.

Parts | Boards

Reset All Filters

Category: All Package: All Temperature: All
Family: All Speed: All

Search:

Click on Boards

Part		IOBs	LUT Elements	FlipFlops	Block RAMs	Ultra RAMs	DSPs	Gb Tra
xc7vx41			257600	515200	880	0	2160	48
xc7vx41			257600	515200	880	0	2160	48
xc7vx415tffv1927-2L	1927	600	257600	515200	880	0	2160	48
xc7vx415tffv1927-1	1927	600	257600	515200	880	0	2160	48
xc7vx485tffg1157-3	1157	600	303600	607200	1030	0	2800	20
xc7vx485tffg1157-2	1157	600	303600	607200	1030	0	2800	20
xc7vx485tffg1157-2L	1157	600	303600	607200	1030	0	2800	20
xc7vx485tffg1157-1	1157	600	303600	607200	1030	0	2800	20
xc7vx485tffg1158-3	1158	350	303600	607200	1030	0	2800	48

? < Back Next > Finish Cancel

rry_Look_Ahead_Adder
at
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cumulator

Tcl Console

New Project Wizard will guide you through the process of selecting design sources and a target device for a new project.

Vivado 2018.3

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Quick Start

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Open Example Project >

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Xilinx Tcl Store >

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New Project

Default Part

Choose a default Xilinx part or board for your project.

Parts | **Boards**

Reset All Filters

Vendor: All Nan

Type Nexys here

Search: NEXYS (5 matches)

Display Name	Preview	Vendor	File Version	Part
Nexys A7-100T		digilentinc.com	1.3	xc7a100tcsg324-1
Nexys A7-50T		digilentinc.com	1.3	xc7a50ticsg324-1L
Nexys4		digilentinc.com	1.1	xc7a100tcsg324-1
Nexys4 DDR		digilentinc.com	1.1	xc7a100tcsg324-1
Nexys Video		digilentinc.com	1.2	xc7a200tsbg484-1

? < Back Next > Finish Cancel

rry_Look_Ahead_Adder
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cumulator

Tcl Console

New Project Wizard will guide you through the process of selecting design sources and a target device for a new project.

Vivado 2018.3

File Flow Tools Window Help Quick Access

VIVADO HLx Editions

XILINX

Quick Start

Create Project >
Open Project >
Open Example Project >

Tasks

Manage IP >
Open Hardware Manager >
Xilinx Tcl Store >

Learning Ce

Documentation and Tutorials
Quick Take Videos >

New Project

Default Part

Choose a default Xilinx part or board for your project.

Parts | **Boards**

Reset All Filters

Vendor: All Name: All

Search: NEXYS (5 matches)

Display Name	Preview	Vendor	File Version	Part
Nexys A7-100T		digilentinc.com	1.3	xc7a100tcsg324-1
Nexys A7-50T		digilentinc.com	1.3	xc7a50ticsg324-1L
Nexys4		digilentinc.com	1.1	xc7a100tcsg324-1
Nexys4 DDR		digilentinc.com	1.1	xc7a100tcsg324-1
Nexys Video		digilentinc.com	1.2	xc7a200tsbg484-1

Choose one based on which ever board you have (don't click on the hyperlink)

Then click Next

< Back Next > Finish Cancel

Tcl Console

New Project Wizard will guide you through the process of selecting design sources and a target device for a new project.

New Project Created

project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

File Edit Flow Tools Reports Window Layout View Help Quick Access Ready

Default Layout

Flow Navigator

- PROJECT MANAGER
  - Settings
  - Add Sources
  - Language Templates
  - IP Catalog
- IP INTEGRATOR
  - Create Block Design
  - Open Block Design
  - Generate Block Design
- SIMULATION
  - Run Simulation
- RTL ANALYSIS
  - Open Elaborated Design
- SYNTHESIS
  - Run Synthesis
  - Open Synthesized Design
- IMPLEMENTATION
  - Run Implementation

PROJECT MANAGER - project_1

Sources

- Design Sources
- Constraints
- Simulation Sources
  - sim_1
- Utility Sources

Hierarchy | Libraries | Compile Order

Properties

Select an object to see properties

Project Summary

Overview | Dashboard

Settings Edit

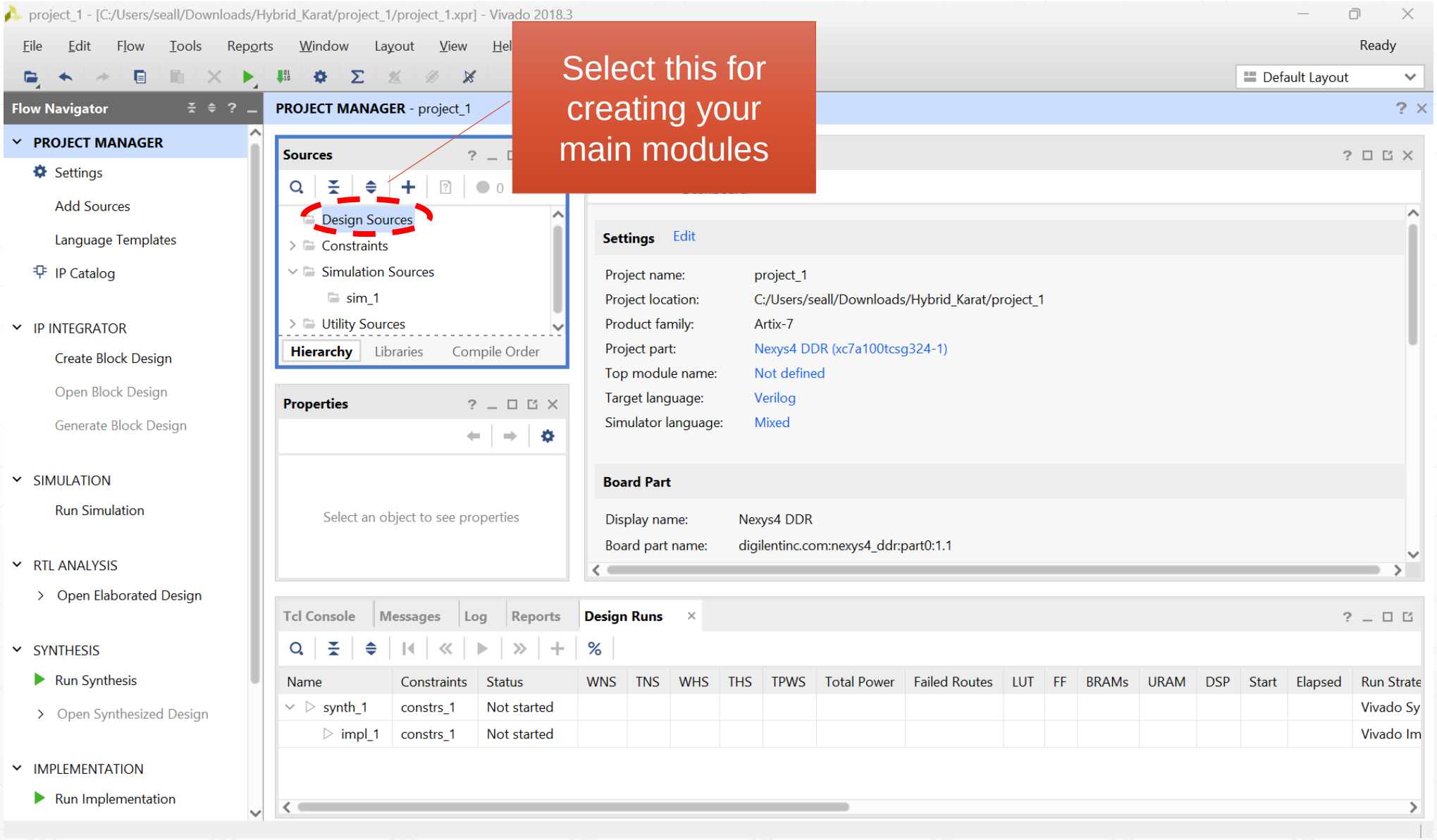
Project name:	project_1
Project location:	C:/Users/seall/Downloads/Hybrid_Karat/project_1
Product family:	Artix-7
Project part:	Nexys4 DDR (xc7a100tcsg324-1)
Top module name:	Not defined
Target language:	Verilog
Simulator language:	Mixed

Board Part

Display name:	Nexys4 DDR
Board part name:	digilentinc.com:nexys4_ddr:part0:1.1

Tcl Console | Messages | Log | Reports | Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAMs	URAM	DSP	Start	Elapsed	Run Strate
synth_1	constrs_1	Not started															Vivado Sy
impl_1	constrs_1	Not started															Vivado Im


project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3
File Edit Flow Tools Reports Window Layout View Hel
Ready
Default Layout
Select this for creating your main modules
Flow Navigator
PROJECT MANAGER
Settings
Add Sources
Language Templates
IP Catalog
IP INTEGRATOR
Create Block Design
Open Block Design
Generate Block Design
SIMULATION
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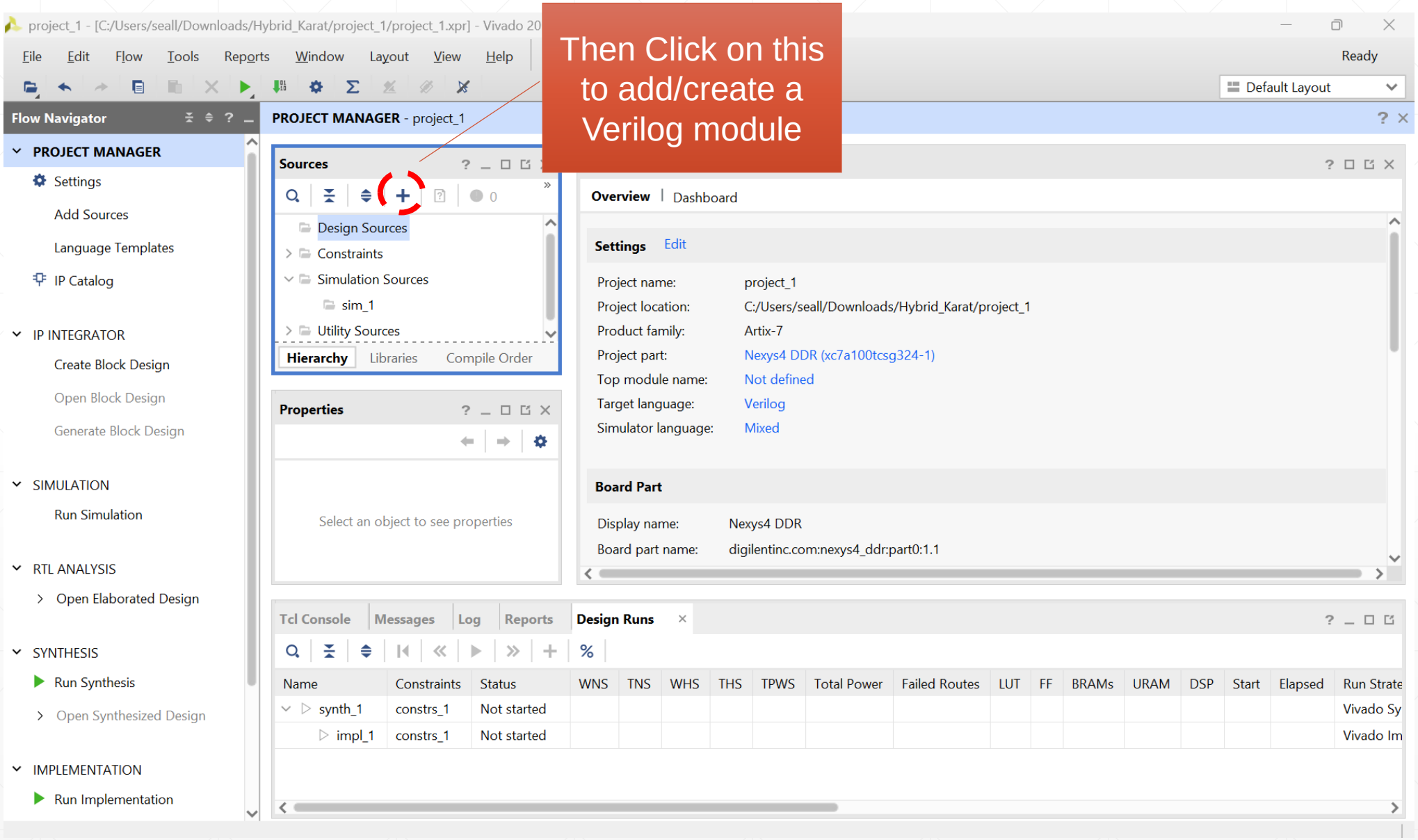
Run Implementation
PROJECT MANAGER - project_1
Sources
Design Sources
Constraints
Simulation Sources
sim_1
Utility Sources
Hierarchy Libraries Compile Order
Properties
Select an object to see properties
Settings Edit
Project name: project_1
Project location: C:/Users/seall/Downloads/Hybrid_Karat/project_1
Product family: Artix-7
Project part: Nexys4 DDR (xc7a100tcsg324-1)
Top module name: Not defined
Target language: Verilog
Simulator language: Mixed
Board Part
Display name: Nexys4 DDR
Board part name: digilentinc.com:nexys4_ddr:part0:1.1
Tcl Console Messages Log Reports Design Runs
Name Constraints Status WNS TNS WHS THS TPWS Total Power Failed Routes LUT FF BRAMs URAM DSP Start Elapsed Run Strate
synth_1 constrs_1 Not started Vivado Sy
impl_1 constrs_1 Not started Vivado Im


project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 20
File Edit Flow Tools Reports Window Layout View Help
Ready
Default Layout
Then Click on this to add/create a Verilog module
Flow Navigator
PROJECT MANAGER
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IP INTEGRATOR
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Open Block Design
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PROJECT MANAGER - project_1
Sources
Design Sources
Constraints
Simulation Sources
sim_1
Utility Sources
Hierarchy Libraries Compile Order
Properties
Select an object to see properties
Overview | Dashboard
Settings Edit
Project name: project_1
Project location: C:/Users/seall/Downloads/Hybrid_Karat/project_1
Product family: Artix-7
Project part: Nexys4 DDR (xc7a100tcsg324-1)
Top module name: Not defined
Target language: Verilog
Simulator language: Mixed
Board Part
Display name: Nexys4 DDR
Board part name: digilentinc.com:nexys4_ddr:part0:1.1
Tcl Console Messages Log Reports Design Runs
Name Constraints Status WNS TNS WHS THS TPWS Total Power Failed Routes LUT FF BRAMs URAM DSP Start Elapsed Run Strate
synth_1 constrs_1 Not started Vivado Sy
impl_1 constrs_1 Not started Vivado Im

project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

File Edit Flow Tools Reports Window Layout View Help Quick Access Ready

Default Layout

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PROJECT MANAGER
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IP INTEGRATOR
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Add Sources

Add or Create Design Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those file types to add to your project. Create a new source file on disk and add it to your project.

Create Source File

Create a new source file and add it to yo project.

File type: Verilog

File name: adder

File location: <Local to Project>

OK Cancel

Type name of the source file and the click on OK

Scan and add RTL include files into project

Copy sources into project

Add sources from subdirectories

< Back Next > Finish Cancel

DSP Start Elapsed Run Strate

Vivado Sy

Vivado Im

Specify and/or create source files to add to the project

project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

File Edit Flow Tools Reports Window Layout View Help Quick Access Ready

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Add Sources

Add or Create Design Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those file types to add to your project. Create a new source file on disk and add it to your project.

Index	Name	Library	Location
1	adder.v	xil_defaultlib	<Local to Project>

Add Files | Add Directories | Create File

Scan and add RTL include files into project

Copy sources into project

Add sources from subdirectories

< Back | Next > | Finish | Cancel

Click on Finish

Specify and/or create source files to add to the project

project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

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Default Layout

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PROJECT MANAGER - project_1

Sources

Design Sc
Constraint
Simulatio
sim_1
Utility Sou

Hierarchy

Project Summary

Properties

Select an

Define Module

Define a module and specify I/O Ports to add to your source file.
For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.

Module Definition

Module name: adder

I/O Port Definitions

Port Name	Direction	Bus	MSB	LSB
	input	☐	0	0

OK Cancel

Click on OK

Tcl Console

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAMs	URAM	DSP	Start	Elapsed	Run Strate
synth_1	constrs_1	Not started															Vivado Sy
impl_1	constrs_1	Not started															Vivado Im

project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

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PROJECT MANAGER - project_1

Sources

Design Sc
Constraint
Simulatio
sim_1
Utility Sou

Hierarchy

Project Summary

Properties

Select an

Define Module

Define a module and specify I/O Ports to add to your source file.
For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.

Module Definition

Module name: ad

I/O Port Definitio

Port Name	Dire			
	input		0	0

OK Cancel

Define Module

The module definition has not been
Are you sure you want to use these

Yes No

Click on Yes

Tcl Console

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAMs	URAM	DSP	Start	Elapsed	Run Strate
synth_1	constrs_1	Not started															Vivado Sy
impl_1	constrs_1	Not started															Vivado Im

project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

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  - Create Block Design
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  - Open Elaborated Design
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  - Open Synthesized Design
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  - Run Implementation

PROJECT MANAGER - project_1

Sources

- Design Sources (1)
  - adder (adder.v)
- Constraints
- Simulation Sources (1)
  - sim_1 (1)

Hierarchy | Libraries | Compile Order

Source File Properties

adder.v

Enabled

Location: C:/Users/seall/Downloads

General | Properties

Project Summary | **adder.v**

C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.srcs/sources_1/new/adder.v

```
12  // Description:
13  //
14  // Dependencies:
15  //
16  // Revision:
17  // Revision 0.01 - File Created
18  // Additional Comments:
19  //
20  //////////////////////////////////////////////////////////////////////////////////
21
22
23  module adder(
24
25      );
26  endmodule
27
```

Tcl Console | Messages | Log | Reports | **Design Runs**

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAMs	URAM	DSP	Start	Elapsed	Run Strate
synth_1	constrs_1	Not started															Vivado Sy
impl_1	constrs_1	Not started															Vivado Im

23:0 Insert Verilog

Write the Module Description

project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

File Edit Flow Tools Reports Window Layout View Help Quick Access Ready

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  - Run Implementation

PROJECT MANAGER - project_1

Sources

- Design Sources (1)
  - adder (adder.v)
- Constraints
- Simulation Sources (1)
  - sim_1 (1)

Hierarchy | Libraries | Compile Order

Source File Properties

adder.v

Enabled

Location: C:/Users/seall/Downloads

General | Properties

Project Summary | adder.v

C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.srcs/sources_1/new/adder.v

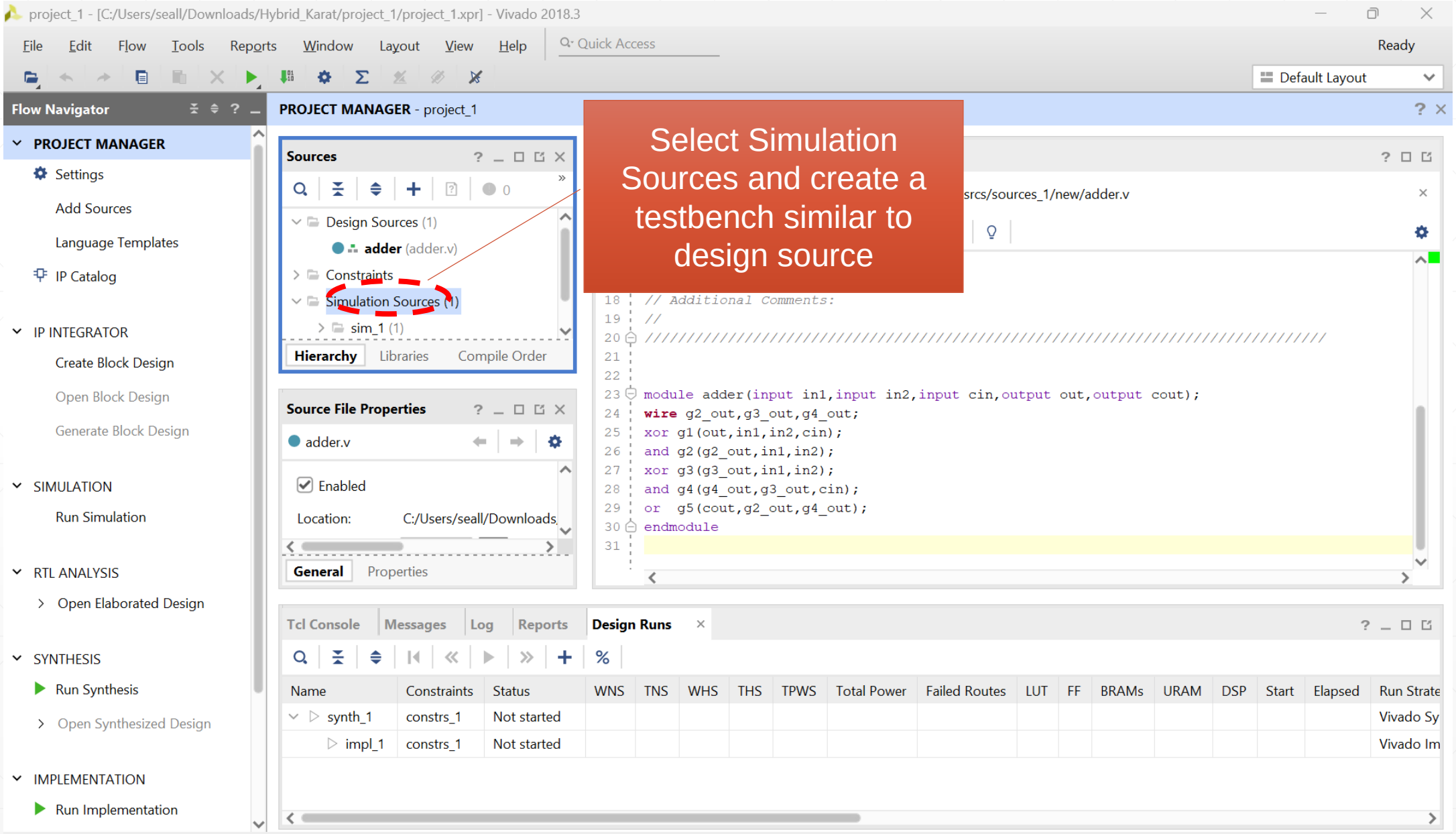
```
16  // Revision:
17  // Revision 0.01 - File Created
18  // Additional Comments:
19  //
20  ////////////////////////////////////////////////////////////////////////////////
21
22
23  module adder(input in1,input in2,input cin,output out,output cout);
24  wire g2_out,g3_out,g4_out;
25  xor g1(out,in1,in2,cin);
26  and g2(g2_out,in1,in2);
27  xor g3(g3_out,in1,in2);
28  and g4(g4_out,g3_out,cin);
29  or  g5(cout,g2_out,g4_out);
30  endmodule
31
```

Tcl Console | Messages | Log | Reports | Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAMs	URAM	DSP	Start	Elapsed	Run Strate
synth_1	constrs_1	Not started															Vivado Sy
impl_1	constrs_1	Not started															Vivado Im

30:9 | Insert | Verilog

Select Simulation Sources



Select Simulation Sources and create a testbench similar to design source

Test bench Created Successfully

project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

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Default Layout

Flow Navigator

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PROJECT MANAGER - project_1

Sources

- Simulation Sources (1)
  - sim_1 (1)
    - test_bench (test_bench.v) (1
      - uut : adder (adder.v)

Hierarchy | Libraries | Compile Order

Source File Properties

test_bench.v

Enabled

Location: C:/Users/seall/Downloads

General | Properties

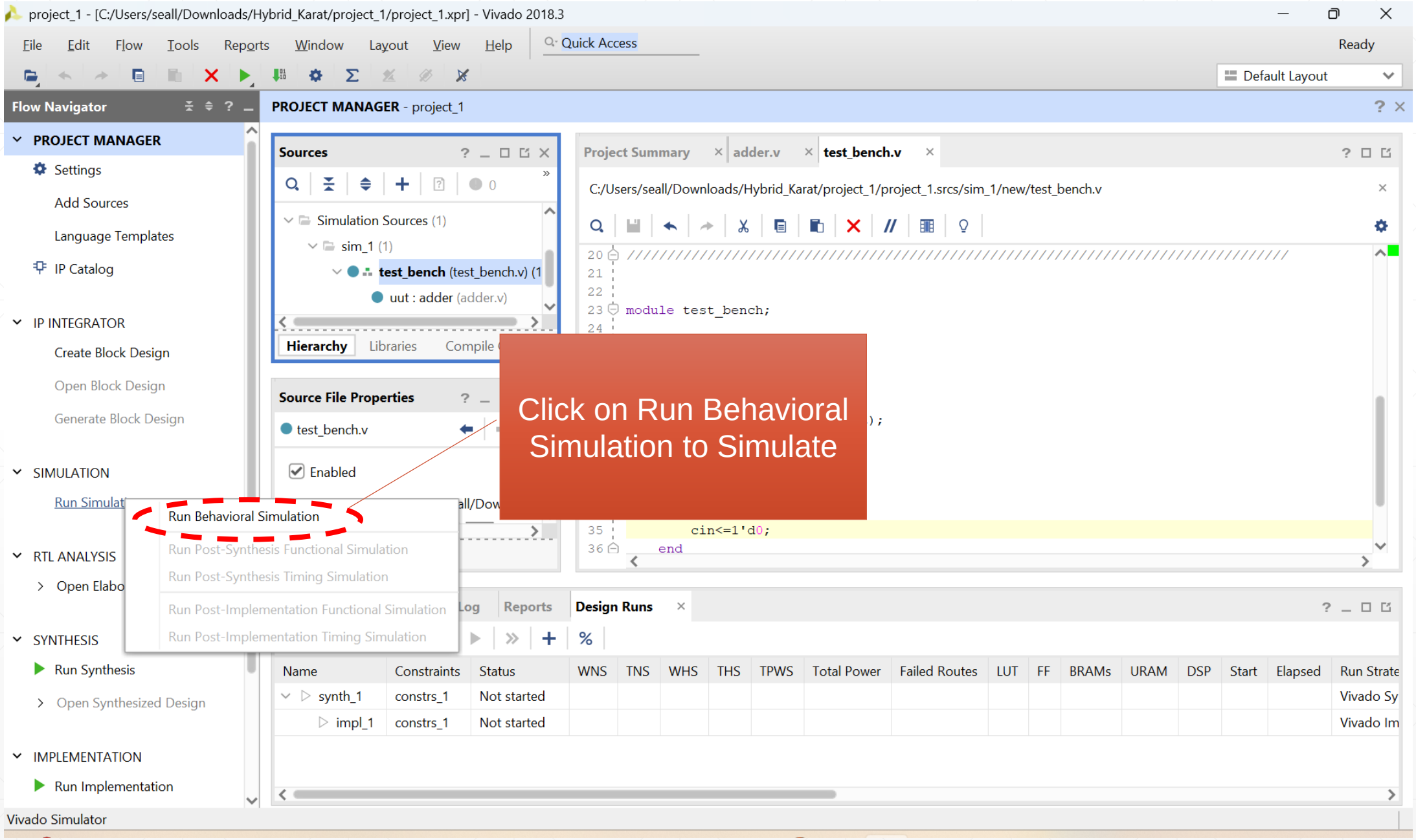
Project Summary | adder.v | test_bench.v

C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.srcs/sim_1/new/test_bench.v

```
20 ////////////////////////////////////////////////////////////////////////////////
21
22
23 module test_bench;
24
25 reg in1, in2, cin;
26
27 wire out, cout;
28
29 adder uut(in1,in2,cin,out,cout);
30
31 initial
32     begin
33         in1<=1'd1;
34         in2<=1'd0;
35         cin<=1'd0;
36     end
```

Tcl Console | Messages | Log | Reports | Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAMs	URAM	DSP	Start	Elapsed	Run Strate
synth_1	constrs_1	Not started															Vivado Sy
impl_1	constrs_1	Not started															Vivado Im


project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3
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PROJECT MANAGER - project_1
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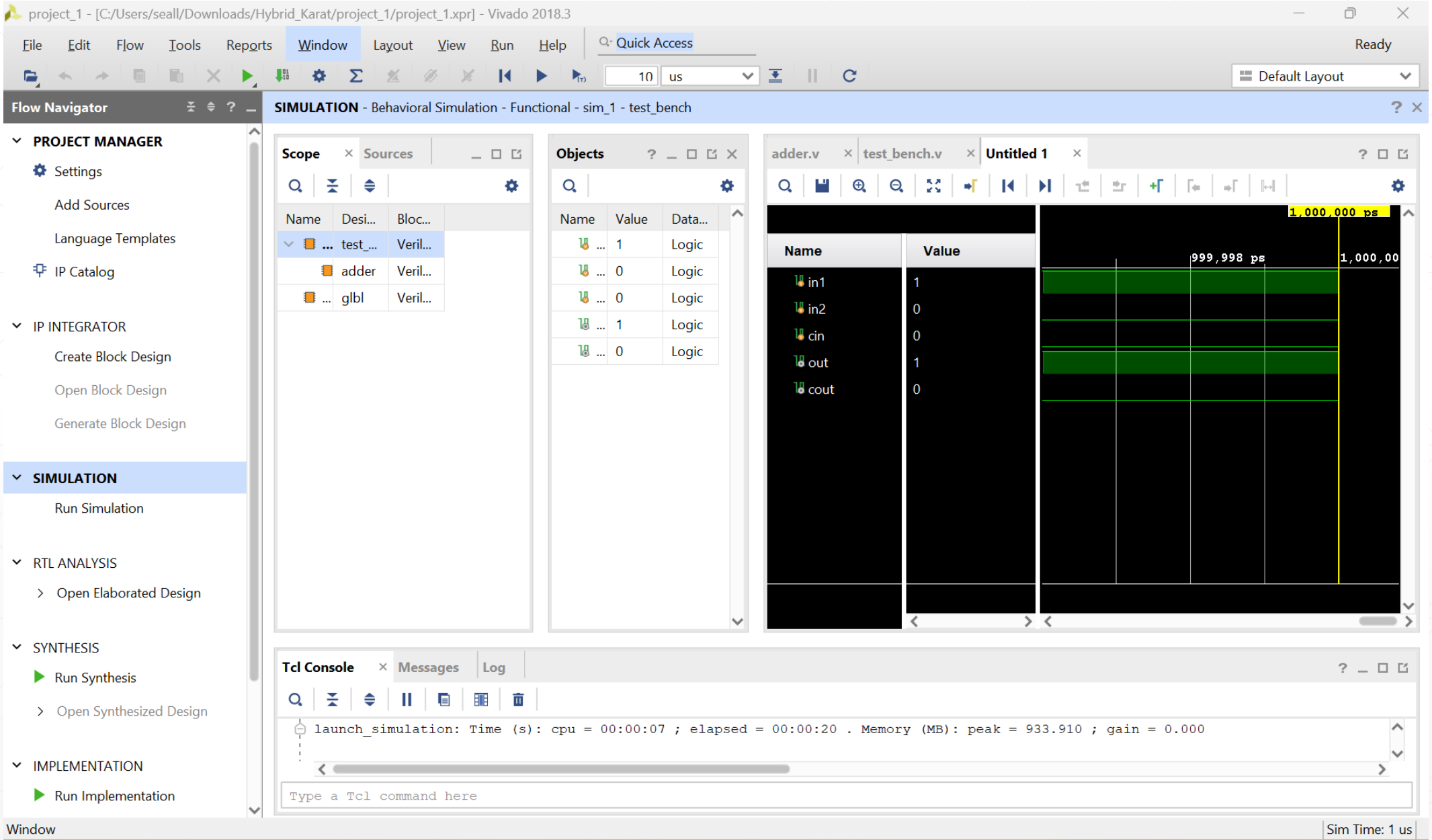
Run Implementation
Sources
Simulation Sources (1)
sim_1 (1)
test_bench (test_bench.v) (1
uut : adder (adder.v)
Hierarchy Libraries Compile
Source File Properties
test_bench.v
Enabled
Run Behavioral Simulation
Run Post-Synthesis Functional Simulation
Run Post-Synthesis Timing Simulation
Run Post-Implementation Functional Simulation
Run Post-Implementation Timing Simulation
Click on Run Behavioral Simulation to Simulate
Project Summary
adder.v
test_bench.v
C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.srcs/sim_1/new/test_bench.v
module test_bench;
cin<=1'd0;
end
Reports
Design Runs
Name Constraints Status WNS TNS WHS THS TPWS Total Power Failed Routes LUT FF BRAMs URAM DSP Start Elapsed Run Strate
synth_1 constrs_1 Not started Vivado Sy
impl_1 constrs_1 Not started Vivado Im
Vivado Simulator

Simulation waveform Window



project_1 - [C:/Users/seall/Downloads/Hybrid_Karat/project_1/project_1.xpr] - Vivado 2018.3

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- Open Synthesized Design

IMPLEMENTATION
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SIMULATION - Behavioral Simulation - Functional - sim_1 - test_bench

Scope | Sources

Name	Design...	Bloc...
test_be...	test_b...	Veril...
uut	adder	Veril...
glbl	glbl	Veril...

Objects

Name	Value	Data.
in1	1	Logic
in2	0	Logic
cin	0	Logic
out	1	Logic
cout	0	Logic
g2_...	0	Logic
g3_...	1	Logic
g4_...	0	Logic

adder.v | test_bench.v | Untitled 1

Name	Value
in1	1
in2	0
cin	0
out	1
cout	0

1,000,000 ps

Select the module of interest from the scope pane. If you want to debug any particular signal, drag and drop the signal/signals form the objects pane onto the waveform window and rerun simulation

Tcl Console | Messages | Log

```
launch_simulation: Time (s): cpu = 00:00:07 ; elapsed = 00:00:20 . Memory (MB): peak = 933.910 ; gain = 0.000
```

Type a Tcl command here

Window Sim Time: 1 us